

DAY 1 — COMPLETE LEARNING LOG

Cyber Security, Operating Systems & Networking Fundamentals

Learning Summary

This Day 1 learning session covered the fundamentals of cyber security, ethical hacking, information security, operating systems, networking, the OSI model, IP addressing and server concepts.

1. Cyber Security Fundamentals

Cyber security is the practice of protecting computers, networks, applications, devices and data from unauthorized access, attacks, damage or disruption.

Topics covered:

- **Ethical Hacking:** authorized security testing to identify vulnerabilities.
- **Information Security:** protecting information and systems through confidentiality, integrity and availability.
- **Threats and Attacks:** understanding common security risks and defensive controls.
- **Antivirus:** software used to detect, prevent and respond to malicious software.

2. Types of Cyber Security

- Physical Security
- Perimeter Security
- Application Security
- Identity and Access Management (IAM)
- Cloud Security
- Data Security
- Email Security
- IoT Security
- Operating System Security
- Hardware Security
- Firmware Security
- Embedded Security
- Container Security
- Network Security

3. Information Security — CIA Triad

- **Confidentiality:** information is accessible only to authorized users.
- **Integrity:** information remains accurate and is protected from unauthorized modification.
- **Availability:** authorized users can access systems and information when required.

4. OSI Model

The OSI model divides network communication into seven layers and helps understand how data moves between devices.

Layer	Description
7. Application	Services used directly by network applications

Layer	Description
6. Presentation	Data formatting, encoding and encryption
5. Session	Establishes and manages communication sessions
4. Transport	End-to-end delivery, reliability and ports
3. Network	Logical addressing and routing
2. Data Link	Frames, MAC addressing and local delivery
1. Physical	Signals, cables and physical transmission

5. Operating System (OS)

An operating system is system software that manages computer hardware and provides services and an environment for applications.

OS topics covered:

- **Process Management:** managing running programs and CPU processes.
- **Service Management:** managing background services.
- **Memory Management:** allocating and managing RAM and virtual memory.
- **Scheduling:** deciding which process receives CPU time.
- **Security Management:** permissions, access control and protection.
- **File System Management:** organizing and accessing files.
- **Server Management:** managing systems that provide network services.
- **Module Handling:** managing OS modules/components.

6. OS Components

- Operating System / Kernel
- Shell / Command Line
- Interpreter / Runtime Environment
- Compiler
- Hardware

7. Operating System Platforms

- Windows
- Linux
- macOS
- Android

8. Networking Fundamentals

Networking is the communication of data between computers and other devices through wired or wireless communication media.

Important concepts covered:

- Network communication and data flow
- OSI model and its seven layers
- Public and private IP addresses

- IPv4 addressing
- Basic network infrastructure concepts

9. IP Addressing

An IP address identifies a device or network interface on an IP network. The notes covered public/private IP concepts and traditional IPv4 address classes.

Class	First Octet	Traditional Purpose
A	1–126	Large networks
B	128–191	Medium networks
C	192–223	Smaller networks
D	224–239	Multicast
E	240–255	Experimental / reserved

10. Servers

A server is a system that provides resources or services to other computers or clients over a network.

- Web Server
- Web/Application Server
- Database Server
- Email Server
- Gaming Server
- Load Balancer
- Proxy Server
- Reverse Proxy Server
- Network Server
- File Server

11. OT / ICS Security

Operational Technology (OT) and Industrial Control Systems (ICS) security focuses on protecting industrial environments such as PLC-based control systems, SCADA and other operational infrastructure.

12. Day 1 Learning Outcome

By the end of Day 1, the learning covered cyber security fundamentals, ethical hacking, information security, CIA triad, cyber security domains, operating system concepts, OS components, networking, the OSI model, IPv4 addressing, server types and an introduction to OT/ICS security.

Suggested GitHub commit: docs: add Day 1 cybersecurity, OS and networking learning log

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